

0 2

In mammals, in the early stages of pregnancy, a developing embryo exchanges substances with its mother via cells in the lining of the uterus. At this stage, there is a high concentration of glycogen in cells lining the uterus.

0 2 . 1

Describe the structure of glycogen.

[2 marks]

0 2 . 2

During early pregnancy, the glycogen in the cells lining the uterus is an important energy source for the embryo.

Suggest how glycogen acts as a source of energy.

Do **not** include transport across membranes in your answer.

[2 marks]

0 2 . 3

Suggest and explain **two** ways the cell-surface membranes of the cells lining the uterus may be adapted to allow rapid transport of nutrients.

[2 marks]

1 _____

2 _____

0 2 . 4

In humans, after the gametes join at fertilisation, every cell of the developing embryo undergoes mitotic divisions before the embryo attaches to the uterus lining.

- The first cell division takes 24 hours.
- The subsequent divisions each take 8 hours.

After 3 days, the embryo has a total volume of $4.2 \times 10^{-3} \text{ mm}^3$.

What is the mean volume of each cell after 3 days? Express your answer in standard form.

Show your working.

[2 marks]

Answer = _____ mm^3



